



α -Synuclein measurements in serum L1CAM-positive extracellular vesicles

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PPMI Project: 204

Summary

Please describe the analysis performed (Should not exceed 200 words)

Unless stated otherwise, PPMI serum samples were blindly processed for immunocapture of L1CAM-positive extracellular vesicles (EVs) using pCBMA-based magnetic beads, followed by MSD measurements. Two vials provided by PPMI were pooled and divided into a 250 μ l aliquots for further downstream processing. In this study, we captured neuronally-derived extracellular vesicles from serum samples of patients with Parkinson's disease carrying a LRRK2 mutation.

Method

Describe the methods used. Sufficient information should be provided to enable investigators to understand the methodology and interpret the data

Beads Conjugation

For conjugation of antibody, the carboxylic acid groups of the pCBMA beads were activated with 2-morpholinoethanesulfonic acid (MES) buffer (15 mM, pH 6.0) containing 50 mg/mL 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC, Sigma) at room temperature for 1 h. Beads were then rinsed with PBS, followed by adding 8 μ g of anti-L1CAM (MA5-14140, Thermo Fisher) per 1 mg beads. The mixture incubated on the rotator overnight at room temperature. The resultant pCBMA-anti-L1CAM beads were washed twice with PBS followed by blocking with protein free blocking buffer + 0.05% Tween-20 and placed on the rotator for 1 h at room temperature.

EV Immunocapture

1. 0.2 mg of pre-coupled beads were added to 0.25 ml of serum sample using 2mL low bind tubes and top up to 1ml using PBS supplemented with protease inhibitors. The mix was incubated overnight at 4°C on the rotator.
2. The beads were then washed once with 500 μ l PBS + 0.05% Tween-20 followed by one wash with PBS. After each washing step the supernatant was removed with the aid of a magnet to hold the beads bound to the tube.





3. The captured EVs were lysed in 30 μ L of 1% Triton X-100 supplemented with protease inhibitors cocktail for 30 min at room temperature on the shaker (900 rpm). The lysate was collected using magnet to remove the beads.

Detection of EV Proteins

Electrochemiluminescence (ECL) was performed in 96-well Meso Scale Discovery (MSD) U-Plex plates that enable multiplexing of markers in the same EV preparation. All steps were performed at room temperature. Two antibodies against CD81 or α -synuclein were used and immobilized in the wells using unique linkers according to the manufacturer's protocol. EV lysate was added in each well and incubated for 2 h at RT. After 3 washes, detection antibodies with Sulfo-TAG labeling were incubated for 1 hour. Following 3 washes and the addition of MSD Read buffer, the plates were read using the MSD-ECL platform (QuickPlex SQ 120), and data were analyzed with the MSD Discovery Workbench 3.0 Data Analysis Toolbox. Antibody pairs for α -synuclein and CD81 were provided by MSD.

Where the CD81 value was below the limit of detection of the assay or below background values, the result is shown as "<LOD", because there was no successful EV capture.

Where the sample was hemolyzed, precluding a reliable readout, the value is shown as "N/A".

For the above reasons, 37 samples were repeated but only after unblinding even though the operator was not aware of the diagnosis while performing the repeat assay. Where relevant the original data are included alongside the updated data from the repeat assay.

For the following subjects, the readouts from both runs did not meet our criteria (either CD81 values were below background levels or the samples were haemolysed). Therefore, we recommend excluding these samples.

ST-00027361, ST-00031492, ST-00027738, ST-00030993, ST-00030985, ST-00029049, ST-00032513, ST-00050461, ST-00050452, ST-00050721, ST-00050460, ST-00050951, ST-00051296, ST-00018747, ST-00050174, ST-00050023, ST-00051149, ST-00030641, ST-00050504, ST-00051289, ST-00050129, ST-00050376, ST-00051072

For the following subjects, we recommended using the replicate 2 value because the generic EV CD81 was higher, suggesting more efficient EV immunocapture and greater confidence in the result:

ST-00027089, ST-00027114, ST-00029668, ST-00030634, ST-00030996, ST-00029043, ST-00050175, ST-00050917, ST-00051666, ST-00051752, ST-00050960, ST-00050994

For the following subjects, we recommended keep both readouts because the generic EV marker (CD81) was detectable and above background values in both runs:

ST-00029696, ST-00051292





References

1. Yan S, Jiang C, Janzen A, et al. Neuronally Derived Extracellular Vesicle α -Synuclein as a Serum Biomarker for Individuals at Risk of Developing Parkinson Disease. JAMA Neurol. 2024;81(1):59-68.
2. Jiang C, Hopfner F, Katsikoudi A, et al. Serum neuronal exosomes predict and differentiate Parkinson's disease from atypical parkinsonism. Journal of Neurology, Neurosurgery & Psychiatry 2020;91:720-729.

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